



Simulation and prediction of vortex-induced vibration of a long suspension bridge using SHM-based digital twin technology

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ABSTRACT

Although wind tunnel tests have already become a design basis for improving the stability of long-span bridges under the action of wind loads, vortex-induced vibration (VIV) events of in-service long-span bridges have still been frequently reported in recent years. The reasons behind this could be attributed to the limitations and uncertainties involved in wind tunnel tests as well as the changes and degradations of in-service bridges. In consideration that most long-span bridges have now been equipped with long-term structural health monitoring (SHM) systems and that digital twins could provide a new solution for simulating and predicting the reality, this study aims at developing a digital twin for VIV of a long suspension bridge to simulate and predict its VIV. The design document-based finite element (FE) model of the bridge is first updated using the measured dynamic characteristics. The numerical model for VIV of the bridge is then established using the vortex-induced force (VIF) model established based on wind tunnel test results. The numerical model together with the VIF model are finally updated against the measured vortex-induced responses (VIR) to produce a digital twin for VIV of the bridge. The relationships between the parameters in the VIF model and wind characteristics are further investigated to enable the digital twin to predict future VIV events. The accuracy of the digital twin for predicting the future VIV and the influence of the VIF parameters obtained from wind tunnel tests on the prediction accuracy both are confirmed through comparisons between the predicted and measured results.

1. Introduction

Wind tunnel tests have become a fundamental aspect of wind-resistant design for long-span bridges to ensure their stability, safety, and functionality under wind loads. To prevent significant vortex-induced vibration (VIV) of the bridge, wind tunnel tests are frequently conducted on the sectional models of the bridge deck to select or confirm an appropriate aerodynamic configuration that can minimize its VIV amplitude. However, VIV events were still reported frequently on in-service long span bridges (Li et al., 2014a; Hwang et al., 2020; Ge et al., 2022) and have risen public concern about the stability and safety of the bridges. The reasons behind this could be attributed to the limitations and uncertainties involved in wind tunnel tests as well as the changes and degradations of in-service bridges. Li et al. (2014b) found that the aerodynamic characteristics of the sectional model are sensitive to Reynolds number and the peak amplitude of VIV increases with increasing Reynolds number. Larsen (1995) found that a low Reynolds

number associated with a small-scale model made the guide vane, which is effective in large-scale tests, failure in preventing vortex formation. Unfortunately, the Reynolds number in wind tunnel tests is often one to two orders of magnitude smaller than that of a full-scale bridge. Structural damping and turbulence intensity of a long-span bridge are also difficult to be quantified in wind tunnel tests during design stage. For example, the Second Severn Crossing cable-stayed bridge exhibited large amplitude VIV when it was open to public, although wind tunnel tests carried out during the design stage concluded that the occurrence probability of large VIV of the bridge would be small. The field measurements conducted late on showed that the structural damping and turbulence intensity on site differed significantly from the assumed values in the wind tunnel tests (Macdonald et al., 2002). Furthermore, the ignorance of the three-dimensional (3D) effects and the imperfect correlation of excitation forces along the bridge deck makes the two-dimensional (2D) sectional model tests underestimate the VIV amplitudes (Zhou et al., 2017).

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On the other hand, to monitor the performance of long-span bridges on site, long-term structural health monitoring (SHM) systems have been installed on the bridges, and some of the SHM systems have already collected wind data and structural responses of the bridges subjected to VIVs. Li et al. (2019) discovered from VIV measurement data that the bridge's aerodynamics during VIVs were subject to time-varying effects caused by nonstationary wind in the field, which differed from VIVs induced by stationary or steady wind in wind tunnels. Liu et al. (2022) proposed a Bayesian inference approach for the identification of vortex-induced force (VIF) parameters using VIV measurement data, in which only the acceleration response data were required but the selection of the frequency band relied heavily on experience. Dan et al. (2022) analyzed the variations of modal frequencies and damping ratios of a suspension bridge under different traffic flow conditions based on the data collected by the SHM system and found that the contribution of traffic loads to the structural damping is significant. The accumulated measurement data also pave the way for employing machine learning (ML) and deep learning (DL) in VIV research. Li et al. (2018) adopted a decision tree that could map the relationship between wind characteristics and VIV modes to automatically identify VIV modes, and they then employed a support vector regression (SVR) to predict the VIV response. They also proposed a basic neural network (NN) to identify the dynamic function during VIV and demonstrated the accuracy of recurrent neural network (RNN) in predicting VIV amplitudes under natural winds (Li et al., 2021).

The concept of digital twins was proposed by Grieves and Vickers (2017) and revisited by the National Aeronautics and Space Administration (NASA) of the United States (US) in 2012. A digital twin is a virtual replica of a physical entity that interacts with the physical entity to provide information-based services throughout its lifecycle (Lin et al., 2021). Four key elements are necessary to achieve this goal: (a) modelling or simulating of a physical entity to create a virtual entity; (b) collecting and merging data from the physical entity; (c) facilitating interaction and collaboration between the physical and virtual entities; and (d) providing services through the virtual entity (Tao et al., 2019). In recent years, digital twins extensively exert their effects on almost all of the phases of industrial production (Liu et al., 2021). For example, Pantoja-Rosero et al. (2023) proposed a damage-augmented digital twin for buildings, in which ML models were used to segment and characterize damages and no manual intervention was needed. For quantitatively assessing and predicting the structural safety of underground pipelines, Li et al. (2022) built a digital twin based on mobile augmented

reality (MAR) and Brillouin fiber optic sensors (BFOSs). In the bridge industry, Shim et al. (2019) developed a bridge maintenance system for prestressed concrete bridges using a 3D digital twin. Lu and Brilakis (2019; 2020) introduced a labeled point clusters-based digital twin for the damage detection of slab and beam-and-slab bridges. Jiang et al. (2021) proposed a digital twin-driven framework for fatigue life prediction of steel bridges. Cheng et al. (2019) built a digital twin for a self-anchored suspension bridge based on the SHM data and found that the fragility curves obtained from the digital twin are closer to the truth than that from the design document-based finite element (FE) model. Lin et al. (2020) established a digital twin for a cable-stayed bridge using the nonlinear model updating method and then used it to assess the bridge's collapse fragility.

Although the data-driven methods mentioned in the second paragraph obtained satisfactory results in predicting VIV, the ignorance of physical knowledge could hinder the understanding of mechanism of VIV events. The work of Lin et al. (2021) proved the excellent capability of digital twins, which could merge the information mining from massive data and physical knowledge, in modelling the physical entity, but most existing digital twins were built for bridge maintenance rather than bridge VIV. In this regard, this study attempts to establish a digital twin for VIV of an in-service suspension bridge to simulate and predict its VIV. The prototype bridge equipped with a SHM system and the VIV events recorded on-site are first introduced in Section 2 to provide data for the digital twin. A high-quality FE model of the bridge is then built in Section 3 to simulate the true state of the in-service bridge through the model updating of the design document-based FE model using the particle swarm optimization (PSO) method. Section 4 establishes a numerical model for VIV of the prototype bridge, in which not only VIF but also static wind forces and buffeting forces are considered to best reflect the reality. Based on the high-quality FE model and the measured acceleration responses during one of the VIV events, the digital twin is developed by optimizing the VIF parameters, initially coming from the wind tunnel tests, using PSO in Section 5. The validation of the digital twin is also conducted in Section 5. In Section 6, the digital twin is employed to predict another VIV event and the comparative results prove the accuracy and capability of the digital twin. The influence of the VIF parameters obtained from wind tunnel tests on the prediction accuracy of the future VIV is also analyzed in Section 6. Finally, the concluding remarks from this study are given in Section 7.



Fig. 1. Location and layout of the Xihoumen Bridge.

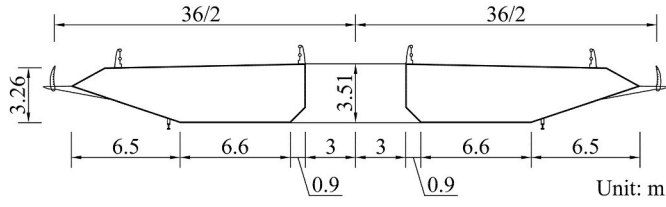


Fig. 2. Cross section of the bridge deck.

2. A long suspension bridge and its VIV events

2.1. A long suspension bridge and its SHM system

The Xihoumen bridge once suffering from VIV is selected as a prototype bridge in this study. The bridge is an asymmetric suspension bridge, located in Zhoushan, China, with a 1650 m main span and a 578 m side span, as shown in Fig. 1. The bridge deck was composed of two steel box girders with a 6 m gap between but connected by crossbeams at a certain interval. The total width of the bridge deck was 36 m and the depth of the deck is 3.51 m. The detailed geometry configuration of the cross section of the bridge deck is presented in Fig. 2. The north and south towers of the bridge, which have a height of 236.5 m, were constructed using reinforced concrete. The main cables of the bridge are comprised of parallel steel wires.

In 2009, a SHM system was implemented on the bridge to monitor both the wind field and structural response. Six Young Model 8100 3D ultrasonic anemometers were installed on the eastern and western lighting columns at the 1/4, 1/2, and 3/4 central spans of the bridge (see Fig. 3), each with a height of 6 m above the upper surface of the deck. The anemometers have a sampling frequency of 20 Hz and only the data recorded by the anemometers at the 1/4 and 3/4 sections of the main span will be used in this study due to those in the middle of main span are very close to the main cable. Twelve force-balance uniaxial accelerometers were installed at the 1/4, 1/2, and 3/4 central spans and 1/2 side spans. Each section had three accelerometers, two were installed on the west side and one on the east side to measure the lateral, vertical, and angular accelerations of the deck. The accelerometers have a sampling frequency of 50 Hz, and their locations are shown in Fig. 4. For more information on the SHM system, refer to Li et al. (2014a).

2.2. VIV events recoded on site

On June 18, 2020, VIV events occurred on the bridge, and the bridge

was closed to traffic for the first time since it was open to the public in 2009. Fig. 5 and Fig. 6 display and contrast the acceleration response time histories of the bridge deck recorded by the accelerometers ACC4 and ACC6 during the time periods from 19:00 to 19:30 (VIV-1) and from 20:00 to 20:30 (VIV-2), in which 1 min time histories in the middle are enlarged to have a clear look of acceleration response. It is seen that the acceleration response time histories of the bridge deck recorded by the accelerometers ACC4 and ACC6 almost overlap, indicating that the VIV events occurred only in the vertical direction. Therefore, only the vertical VIVs of the bridge need to be considered in this study. Moreover, in this study the VIV-2 event will be used to establish the digital twin because vortex-induced response (VIR) in this event was relatively stable compared with VIV-1, whereas the VIV-1 event will be employed to examine the accuracy of the established digital twin. Apart from the VIV-1 and VIV-2, other VIVs occurring on that day cannot be used because high quality wind records were not available somehow.

The wind measurement data collected by the anemometers installed at the 1/4 span of the bridge deck is analyzed, and the wind rose diagrams are displayed in Fig. 7 for the VIV-1 and VIV-2 events. It is seen that the incoming wind directions during the two events were from southeast direction and nearly perpendicular to the bridge deck. The wind measurement data collected by the anemometer UA1, which was located on the windward side, will be therefore used for the subsequent analysis.

3. FE model and model updating of the bridge

Because VIV often occurs at only one of the modes of vibration of a bridge, the mode decomposition method is usually adopted to analyze the VIV of the bridge and accordingly the VIV simulation/prediction relies on the bridge's modal information including mode shapes and natural frequencies. Therefore, a high-quality FE model that can provide the modal information with high accuracy is essential for simulating/predicting VIV. However, the uncertainties in the FE model, such as material properties, boundary conditions, structural damage or others, can lead to discrepancies in modal information between the design document-based FE model and the actual bridge. Updating the FE model according to field-measured vibrations is thus necessary before VIV simulation/prediction.

3.1. The design-document-based FE model

The design-document-based FE model of the bridge was established via ANSYS 2021 R1. BEAM 4 was employed to model most of the bridge

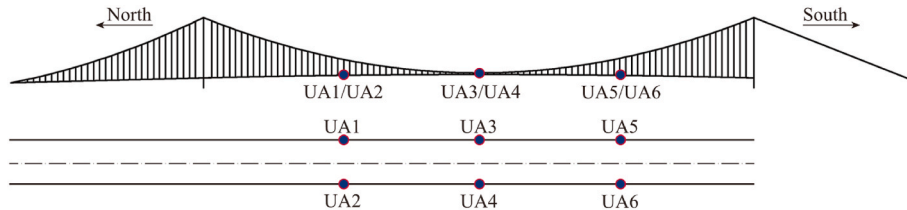


Fig. 3. Anemometer locations used in the SHM system (Li et al., 2014a).

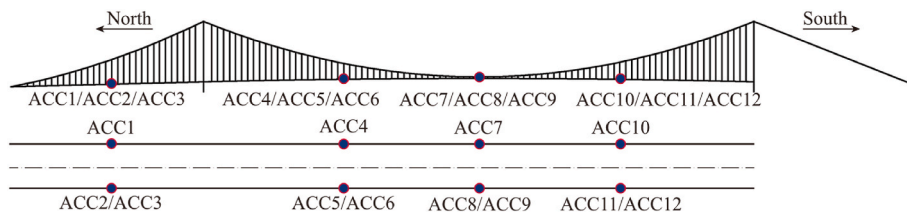


Fig. 4. Accelerometer locations used in the SHM system (Li et al., 2014a).

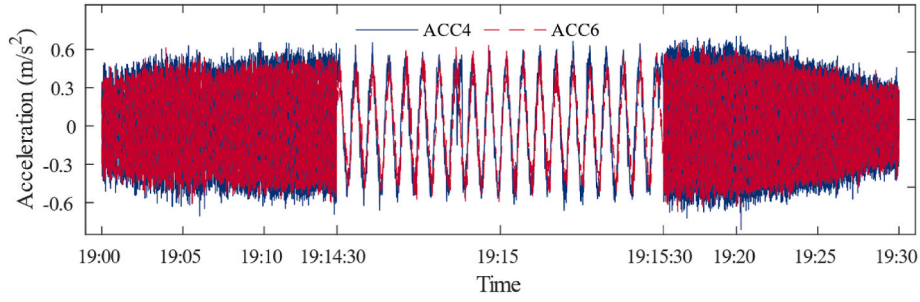


Fig. 5. The acceleration response time histories of the deck recorded by ACC4 and ACC6 during VIV-1.

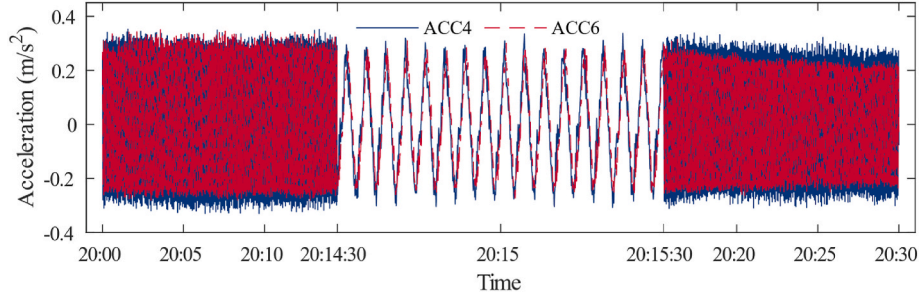


Fig. 6. The acceleration response time histories of the deck recorded by ACC4 and ACC6 during VIV-2.

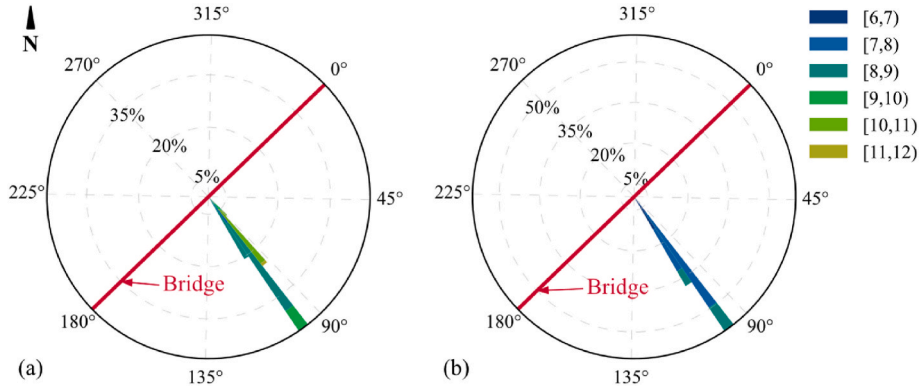


Fig. 7. Wind rose diagrams for the 1/4 span of the bridge deck: (a): VIV-1; (b): VIV-2.

components including the deck, the piers as well as the towers. LINK 10 with initial strain was used to simulate the cables while the ancillary facilities added to the deck were modelled by MASS 21 elements. The first six vertical mode shapes from the eigenvalue analysis of the design-document-based FE model are plotted in Fig. 8.

3.2. FE model updating

The commonly used framework of FE model updating involves three major components: (a) set up an objective function, (b) select updating parameters, and (c) perform optimization algorithm (Lin et al., 2020).

3.2.1. Objective function

It is crucial to establish an appropriate objective function that accurately reflects the differences between the simulated and measured quantities (indices). The objective function should consider not only the sensitivity of the indices to the structural parameters but also the quality of the indices obtained from the measured data. Various types of indices have been utilized in previous studies to create objective functions for FE model updating, including modal information (Benedettini and Gentile, 2011; Zhu et al., 2015; Xiao et al., 2015; Wang et al., 2018), influence

lines (Zhu et al., 2015; Xiao et al., 2015; Wang et al., 2018), response time histories (Gu et al., 2012; Xin et al., 2019), and local responses (Wang et al., 2017, 2018; Schlune et al., 2009), among others. In consideration of the limited type and number of accelerometers installed on the concerned bridge, the commonly used natural frequencies are used as the indices in the present model updating. The fast Bayesian fast Fourier transform (FFT) method (Au, 2011) is employed to identify the natural frequencies of the in-service bridge from the measured acceleration response time histories between 16:00 and 18:00 on June 18, 2020 (see Fig. 9), during which the bridge vibrated due to the ambient excitation rather than VIF. The summation of the normalized errors between the simulated natural frequencies and the measured ones is then used to construct the objective function as

$$OBJ(x) = \sum_{i=1}^n [(f_i(x) - f_{i,exp})/f_{i,exp}]^2 \quad (1)$$

where x is the vector formed by the selected updating parameters, which will be introduced in the following subsection; $f_i(x)$ and $f_{i,exp}$ are the i th natural frequencies obtained from the FE model and the measurement data, respectively; and n is the number of the natural frequencies

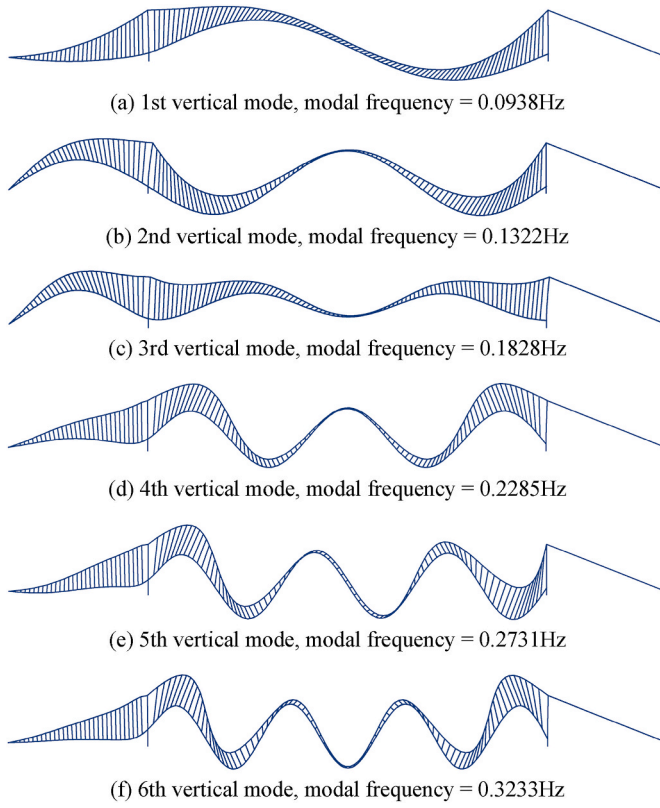


Fig. 8. The first six vertical mode shapes from the design-document-based FE model.

considered in the model updating, which is 6 in this study.

3.2.2. Updating parameters

The selection of parameters for model updating is a crucial step. Choosing incorrect parameters can result in a model that fails to accurately reproduce the required dynamic properties or accurately represent the true structure. Additionally, the number of parameters selected affects the complexity of the optimization problem and the computational resources required. To reduce computational demands and enable successful model updating, sensitivity analyses should be conducted to identify parameters that are uncertain but sensitive to the indices used in the objective function. Based on the results of the sensitivity analysis, four parameters have been selected for updating: the elastic module of the deck (E_{deck}), the material density of the deck (ρ_{deck}), the material density of the main cables (ρ_{cable}), and the initial forces of the main cables (F_{cable}).

3.2.3. Optimization algorithm

An optimization algorithm is used to minimize the objective function by adjusting the updating parameters. There are quite a few different

methods for solving an optimization problem, such as PSO (Kennedy and Eberhart, 1995), genetic algorithm (GA) (Rocke, 2000), and simulated annealing (SA) (Kirkpatrick et al., 1983). In this study, PSO is adopted. PSO involves representing each potential solution as a “particle” in a D-dimensional space, where D is the number of parameters being optimized. The swarm of particles moves through the parameter (search) space, influenced by their own and their neighbors’ performance as well as their inertia, similar to the behavior of bird flocks and fish schools, as depicted in Fig. 10.

In contrast to other nature-inspired algorithms, PSO relies on cooperation and competition among individuals across generations (iterations). The exchange of information among particles is a crucial aspect of the algorithm and can be limited to a local neighborhood (partial PSO) or extended to the entire swarm (global PSO), as noted by Marini and Walczak (2015). By mapping the objective function to the performance of the particles and the updating parameters to the positions of the particles, PSO can easily exert its effects in the FE model updating.

3.3. The updated FE model

Note that the performance evaluations of the particles, based on the analyses of the FE model using the parameters related to their positions, in each iteration are independent of each other. Thus, the parallel calculation provided by MATLAB R2022a (academic use) can be used to distribute the intensive workloads to all the 32 cores of the AMD Ryzen Threadripper 3970X processor in this study to accelerate the updating process. The design-document-based parameters and their updated values are listed in Table 1 while the natural frequencies of the first six vertical modes of vibration of the in-service bridge as well as those from the FE models are listed in Table 2. It can be found from Table 2 that not only the differences in the natural frequencies between the design document-based FE model and the updated FE model are small, but also the differences between the measured frequencies and those from the FE models are small. This is because the concerned bridge is still new and the design document-based FE model is of high quality. Nevertheless, FE model updating is often required when developing the digital twin for other bridges, especially for the bridges used for a long time with accumulated structural damage. This study thus includes FE model updating to demonstrate the entire process for developing the digital

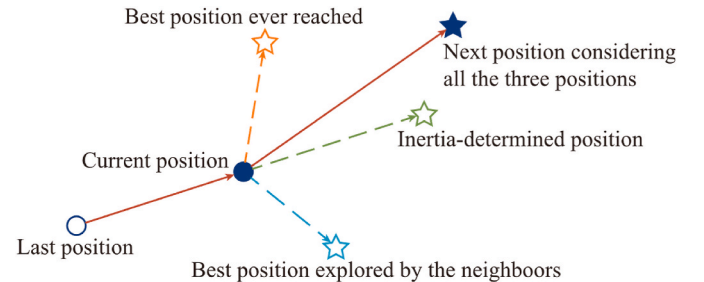


Fig. 10. Position updating strategy of particles in PSO.

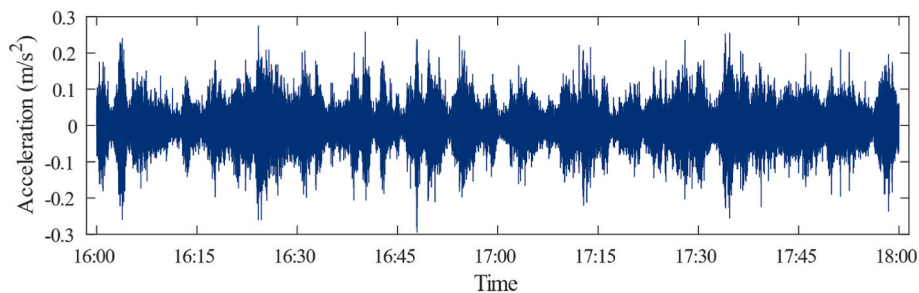


Fig. 9. Ambient-induced acceleration responses of the deck (16:00 to 18:00 June 18, 2020).

Table 1

Design-document-based and updated parameters.

Parameter name	Document-based	Updated	Difference (%)
elastic module of the deck (E_{deck})	210.0GPa	226.6GPa	7.90
material density of the deck (ρ_{deck})	8242.9Kg/m ³	8695.3Kg/m ³	5.49
material density of the main cables (ρ_{cable})	8242.7Kg/m ³	7571.8Kg/m ³	-8.14
initial forces of the main cables (F_{cable})	2.71×10^5 kN	2.78×10^5 kN	2.58

twin. The updated FE model will be used for establishing the digital twin for VIV of the bridge.

4. Numerical simulation of VIV

Based on the updated FE model, the numerical model for simulating VIV of the bridge shall be developed before a digital twin can be established. The development of an accurate and efficient numerical model shall try to balance the enormous computation burden brought by complex mathematical equations with the unacceptable errors coming with unreasonable assumptions.

4.1. Modelling of wind loads on the bridge during VIV

In consideration that the concerned prototype bridge was close to the traffic during the VIV events on June 18, 2020, only wind loads acting on the bridge need to be considered in developing the numerical model. Additionally, since VIV of the bridge deck only occurred in the vertical direction, the numerical model only needs to consider wind loads in the vertical direction. Wind loads acting on the bridge in a turbulent wind field can be divided into three components: static wind forces, buffeting forces, and VIF, which are caused by the mean wind, turbulence component, and aeroelastic interaction between wind and bridge motion, respectively.

For VIF, the pure-vortex-induced forces are negligible when the VIV of the bridge deck enters its steady stage according to the works of Ehsan and Scanlan (1990) and Zhu et al. (2017a). Therefore, VIF can be classified as self-excited forces, for all the remaining terms in the VIF models (Zhu et al., 2013, 2017a; Simiu and Scanlan, 1996) are induced by interaction between incoming wind and bridge motion. Several semi-empirical models for VIF have been developed in terms of the wind tunnel test results (Simiu and Scanlan, 1996; Ehsan and Scanlan, 1990; Larsen, 1995), but most of them concern smooth winds only. Zhu et al. (2013, 2017a) recently proposed a VIF model applicable in a turbulent wind field. In this model, the VIF acting on a twin-box deck section in a turbulent wind field in the vertical direction can be expressed by

$$F_{vi} = \frac{\rho D^2}{m} [Y_1 (1 - \varepsilon \dot{\eta}^2 - \varepsilon_t w_r^2) \dot{\eta} + Y_2 \eta] \quad (2)$$

where Y_1 , Y_2 , and ε are the parameters to-be-determined by wind tunnel tests under smooth incoming winds; ε_t is the to-be-determined param-

eter related to turbulent wind-induced damping term; $w_r = \frac{w}{U}$ is the dimensionless vertical turbulence component, where w and U are the vertical fluctuating wind speed and the horizontal mean wind speed, respectively; D is the depth of the deck; m is the mass per length of the deck; ρ is the air density; $\eta = y/D$ is the dimensionless vertical displacement of the deck, where y is the vertical displacement of the deck. The span-wise correlation of VIF was also investigated by Zhu et al. (2017b), and the relevant results are also used in the numerical model.

The buffeting forces acting on the bridge deck in the vertical direction can be expressed according to the quasi-steady theory (Scanlan and Jones, 1990).

$$F_{bu} = \frac{1}{2} \rho U^2 B \left[2C_{BL} \frac{u}{U} \gamma_1 + \left(C'_{BL} + \frac{D}{B} C_{BD} \right) \frac{w}{U} \gamma_2 \right] \quad (3)$$

where C'_{BL} is the slope of C_{BL} ; C_{BL} and C_{BD} are the lift and drag coefficients, respectively; u is the turbulent wind in the mean wind direction; γ_1 and γ_2 are the aerodynamic admittance functions that are conservatively taken as unit; and B is the width of the deck.

The static wind forces acting on the bridge deck include three components: drag, lift and moments (Simiu and Scanlan, 1996). The lift component is concerned in this study and it can be expressed by

$$F_{st} = \frac{1}{2} \rho U^2 B C_{BL} \quad (4)$$

4.2. Equation of motion of the bridge

Based on the updated FE model of the entire bridge, the equation of motion of the vortex-excited bridge deck in a turbulent wind in the vertical direction can be assembled and written as

$$M_b \ddot{Z}_b + C_b \dot{Z}_b + K_b Z_b = F_{vi} + F_{bu} + F_{st} \quad (5)$$

where M_b , C_b , and K_b are the mass, damping, and stiffness matrices of the bridge deck; Z_b is the vector of vertical displacement of the bridge deck; F_{vi} , F_{bu} , F_{st} are the vectors of the VIFs, buffeting forces and static forces acting on the bridge deck.

In consideration that the VIV event of the bridge deck occurred in a single vertical mode of vibration, the modal superposition method can be used and the vertical displacement of the deck at time t can be expressed as

$$Z_b(t) = \varphi_n q(t) \quad (6)$$

where φ_n is the n -order mode shape of the deck; and $q(t)$ is the generalized coordinate for the n -order mode of vibration of the deck.

With the Rayleigh damping assumption and by applying Eq. (6) to Eq. (5), the generalized equation of motion of the bridge deck for the n -order mode of vibration can be obtained as

$$M_n (\ddot{q}(t) + 2\omega_n \xi_n \dot{q}(t) + \omega_n^2 q(t)) = \int_0^L (F_{vi} + F_{bu} + F_{st}) \varphi_n dx \quad (7)$$

where L is the deck length; $M_n = \varphi_n^T M_b \varphi_n$ represents the n -order generalized mass of the bridge; ω_n and ξ_n are the n -order circular frequency and damping ratio of the bridge, respectively.

Table 2

Natural frequencies of the prototype bridge and the FE model before and after the updating.

Mode No.	Measured frequency (Hz)	Measured damping ratio (%)	Design document-based FE model		Updated FE model	
			Frequency (Hz)	Difference (%)	Frequency (Hz)	Difference (%)
1	0.0946	0.66	0.0938	-0.85	0.0943	-0.32
2	0.1321	1.06	0.1322	0.08	0.1326	0.38
3	0.1835	0.18	0.1828	-0.38	0.1835	0.00
4	0.2290	0.12	0.2285	-0.22	0.2295	0.22
5	0.2756	0.16	0.2731	-0.91	0.2745	-0.40
6	0.3252	0.11	0.3233	-0.58	0.3253	0.03

The damping ratios of the first six vertical modes of vibration are set as those identified from the measured ambient-induced acceleration responses (see Table 2). The Newmark-Beta method is employed to solve the equation.

4.3. The parameters in the VIF model from wind tunnel tests

Before using the above numerical model to simulate VIV of the bridge deck and to develop a digital twin, a set of initial parameters in the VIF model, which can reflect the VIV characteristics of the prototype bridge as close as possible, shall be decided. This can be realized by using the VIF parameters obtained from the design document-based wind tunnel tests. For the prototype bridge concerned, Zhu et al. (2018) carried out extensive wind tunnel tests of VIV of the deck section and identified the VIF parameters, as listed in Table 3. In these parameters, Y_1 , Y_2 and ε were fitted from the VIF time histories in a smooth flow but ε_t was fitted from the VIF time histories in different turbulent winds. It is noted that although ε_t relates to vertical wind turbulence, as shown in Eq. (2), Zhu et al. (2018) take the same value, 115.1, to simulate VIV in turbulent winds with turbulence intensities ranging from 2% to 7% to simplify the problem. This may be one of the reasons why the wind tunnel test parameters are not fully able to simulate on-site (observed) VIV accurately and why the digital twin needs to be developed, as discussed subsequently in Section 6. The aerodynamic coefficients of the deck section were also obtained through the wind tunnel tests (Ge, 2011) with $C_{BD} = 0.9835$, $C_{BL} = -0.0327$ and $C'_{BL} = 1.9796$. Compared with VIF parameters, these aerodynamic coefficients related to the buffeting forces and static forces are relatively stable and reliable.

5. Development of a digital twin

5.1. Procedure for developing a digital twin

To develop a digital twin for simulating and predicting VIV of the prototype bridge, the VIF parameters in the numerical model for simulating VIV of the prototype bridge shall be identified (or updated) against the measured acceleration responses of the prototype bridge during VIV events so that the updated numerical model, that is, the digital twin, can simulate and predict VIV events of the prototype bridge in the future. The objective function used in this updating is to identify the best VIF parameters to make the differences between the predicted and measured VIRs as small as possible. The sum of the weighted squares of the differences between the two acceleration response time histories, as shown by Eq. (8), is adopted as an objective function in the optimization process.

$$DIF = \frac{1}{n} \sum_{i=0}^n [Weight \times (A_{pre} - A_{mea})]^2 \quad (8)$$

where DIF represents the differences between the predicted and measured acceleration responses; $Weight$ is a weighting vector whose

elements decrease from 100 to 1 along the length of the acceleration response time histories; A_{pre} and A_{mea} are the predicted and measured acceleration response time histories, respectively; and n is the number of the sampling points in the acceleration response time histories.

The $Weight$ adopted here is a weighting vector having the same dimension as the data time series. Large values are assigned to the front elements of the vector while small values are assigned to the back elements so that the front elements of the data time series will be weighted more than the back elements of the data time series. This weighting measure makes the optimization method pay more attention to the front elements of the data time series and accelerate the optimization, as the current state of the coupled dynamic system is determined by not only the current excitation but also the previous state. When it comes to the optimization method, the unconstrained PSO method, in which the limits for the particle positions are set as infinity, is adopted as it is hard to estimate the discrepancies between the VIF parameters from the wind tunnel tests and the actual ones for the prototype bridges.

With the selected objection function, updating parameters, and optimization method for the numerical model updating as well as the prototype bridge and its SHM system and updated FE model, the integrated procedure for developing the digital twin for simulating VIV of the prototype bridge can be summarized in Fig. 11.

5.2. Identification of VIF parameters

In order to identify the VIF parameters in the numerical model based on the measured acceleration responses of the bridge deck during VIV events, an efficient computing method is required to handle the intensive computational workload resulting from the repetitive time history analyses of the numerical model during the updating process. To address this issue, the parallel computing model updating method has been employed in this study to accelerate and enable the numerical model updating process.

Fig. 12 illustrates the procedure for identifying the VIF parameters in the numerical model against the acceleration responses measured from the prototype bridge during the VIV events. Firstly, the initial VIF parameters obtained from the wind tunnel tests are disturbed randomly to initialize a unique VIV simulation model for each worker, an individual core in the 32-core central processing unit (CPU). The parallel pool provided by MATLAB R2022a (academic use) is used to allocate the models to the workers. The workers then solve their models according to the wind speed time histories measured by the anemometer and contrast the predicted acceleration responses with the measured acceleration responses to evaluate the optimization performance in terms of DIF defined by Eq. (8). To accelerate the optimization procedure, the actual response state (x_0, v_0, a_0) of the bridge at the initial time (t_0) is assigned to each worker as the initial state. The measured state needs to be transformed to the generalized coordinate before the assignment. As only acceleration responses are available from the prototype bridge, the integration with detrending in the time domain is adopted to obtain the time histories of velocity and displacement responses. After that, the unconstrained PSO is employed to update the VIF parameters of each worker for the next iteration according to the worker's performance, inertia as well as the neighbors' performance. This procedure continues until the maximum iteration number is reached or the worker performances cannot be further promoted. The best VIF parameters can be finally acquired from the worker reaching the lowest DIF . It is worth of emphasizing that the maximum iteration number is set as large as possible in this study to avoid the early stop, but infinite iterations resulting from errors can be found and terminated during the optimization.

To consider the variations of mean wind speed, turbulence intensity, and deck vibration amplitude during the VIV event, this study divides wind speed and acceleration response time histories minute by minute with 55 s overlap and wind characteristics are counted within each minute rather than 10 min. Fig. 13 displays the performances of the

Table 3

VIF parameters obtained from wind tunnel tests (Zhu et al., 2018).

Reduced wind speed	Y_1	Y_2	ε	ε_t
7.901	1.63	3.998	85.7	115.1
8.379	9.95	4.801	161.0	115.1
8.828	35.10	4.645	285.2	115.1
9.280	50.02	4.726	351.7	115.1
9.740	48.50	4.983	422.6	115.1
10.199	41.10	5.445	593.4	115.1
10.667	30.65	5.883	949.8	115.1
11.136	18.24	6.520	1671.0	115.1
11.569	1.74	7.104	8464.3	115.1
12.046	12.50	8.726	4817.0	115.1

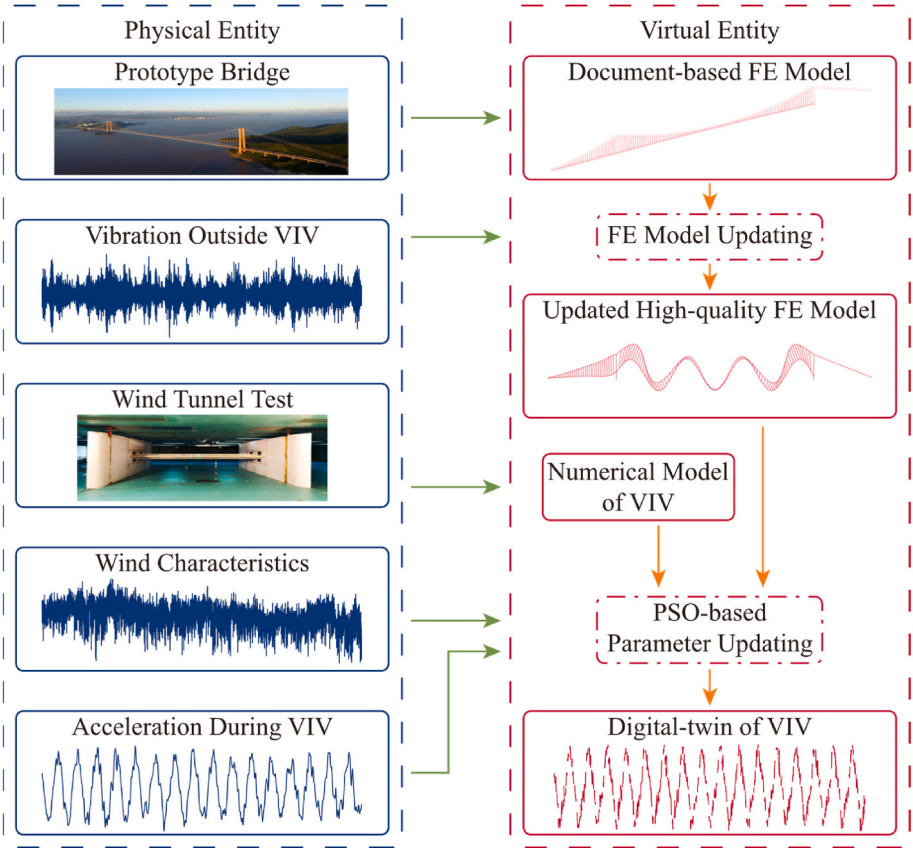


Fig. 11. Procedure for developing a digital twin to simulate VIV of the prototype bridge.

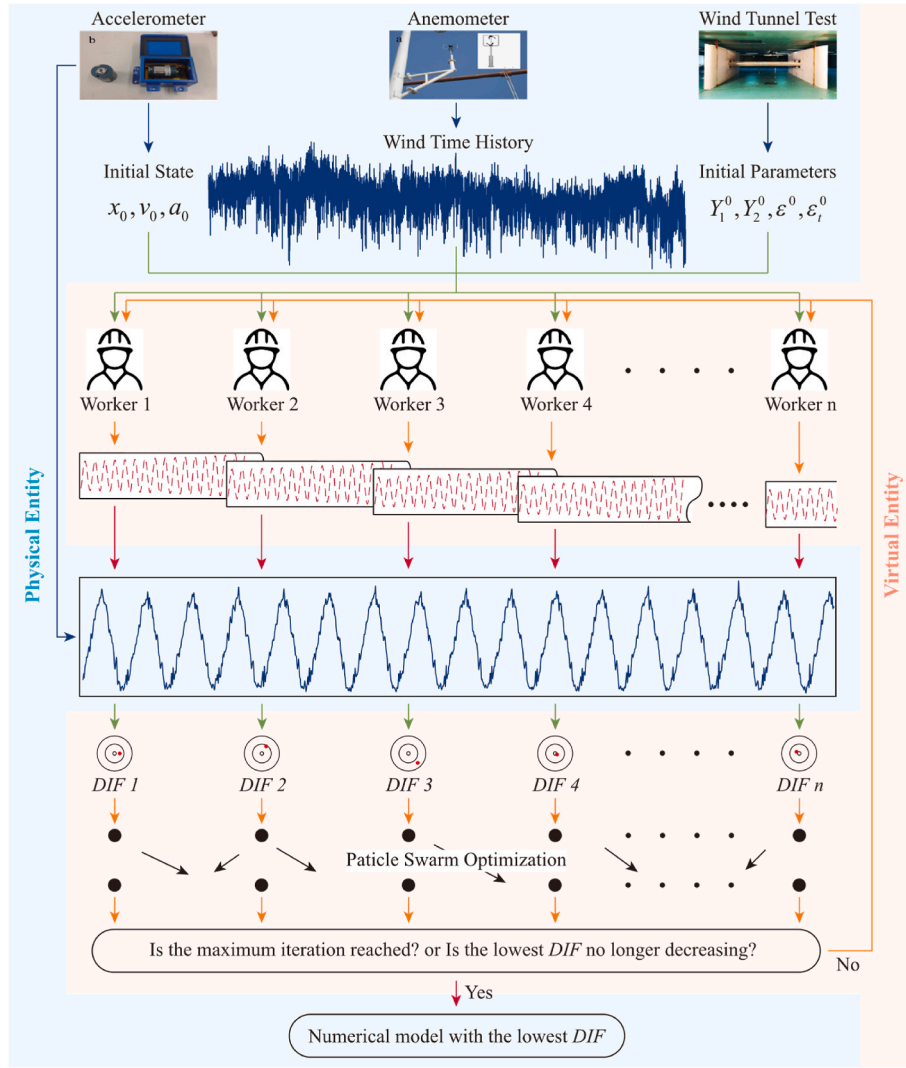


Fig. 12. A parallel computing model updating method for identifying VIF parameters.

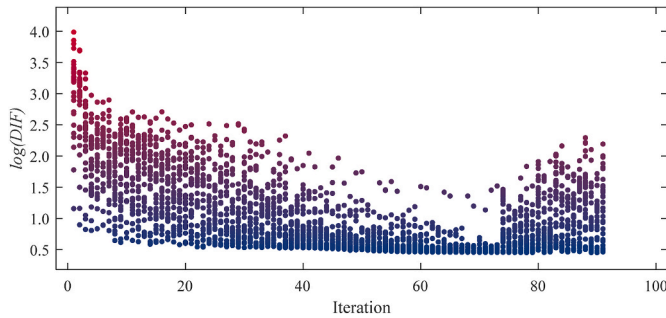
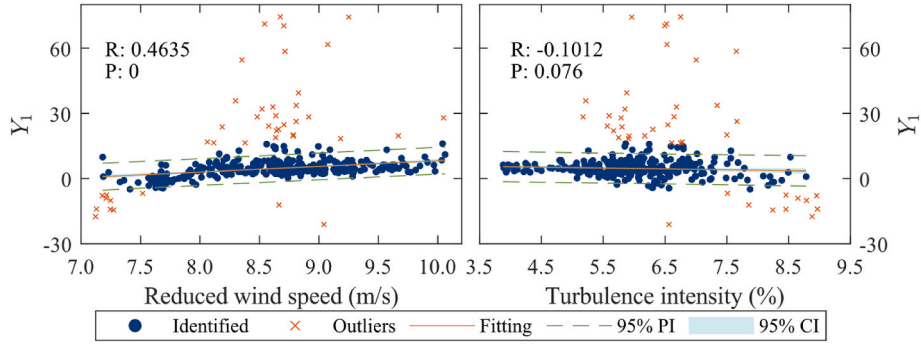
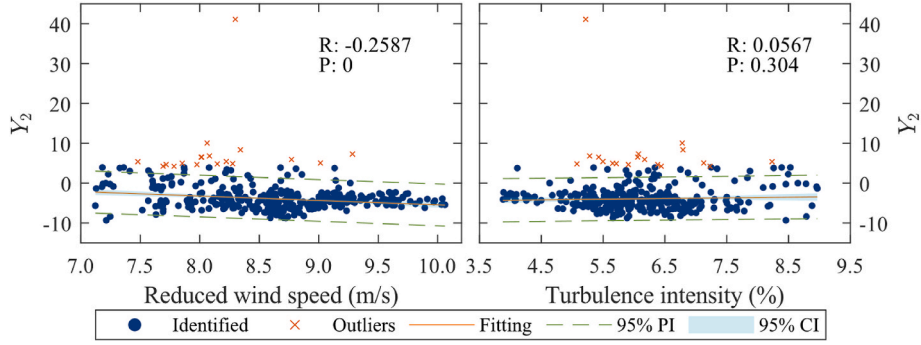
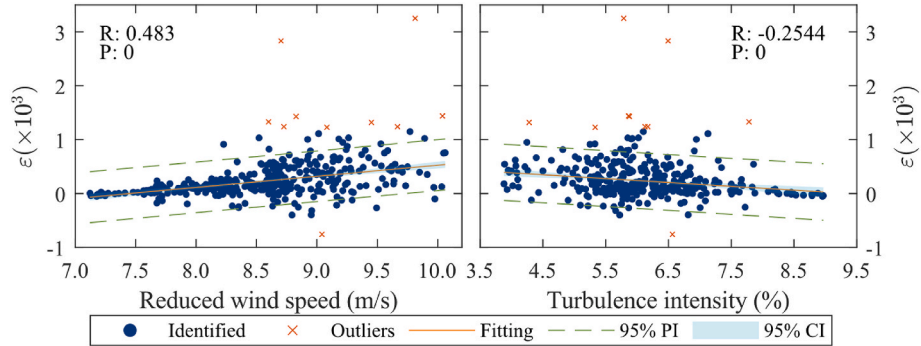


Fig. 13. Performance of the particles in identifying VIF parameters (20:04 to 20:05).

workers for identifying the VIF parameters during 20:04 to 20:05. Different from other applications of PSO, which usually set a pre-defined target for the particles as a threshold to stop the iteration, this study controls the iteration by monitoring the fluctuation of the worker performances since an all-time-available pre-defined target is hard to be found. This is because the noises existing in the measurement signals make it impossible to perfectly contrast the measured and predicted acceleration responses. Therefore, this study stops the iteration when the best performance no longer decreases (the change between two

adjacent iterations is smaller than 0.01) in the last 50 iterations. To avoid locally optimal solutions, random disturbance is added to the workers when their performances tend to be steady, and the degree of the disturbance ascends with the accumulation of steady iterations but does not exceed 20% from the position determined by the position updating strategy. The jump after 70 iterations in Fig. 13 indicates the action of the disturbance and no descending of the lowest value proves that the best parameters have been found.

The identified VIF parameters, the fitting lines, and their 95% confidence interval (CI) as well as 95% prediction interval (PI) are displayed in Fig. 14 to Fig. 17 as the functions of the reduced wind speed and turbulence intensity for the prototype bridge during the VIV-2 from 20:00 to 20:30. The parameters in the fitting lines are listed in Table 4. The Pearson correlation coefficients (R) and corresponding P values (P) are also marked in the figures. Some outliers, defined as the points that are more than double quartiles larger than the upper quartile (75%) or smaller than the lower quartile (25%), are marked by "X" in the figures and are not used in the following analysis. These figures show that both the aerodynamic damping term (Y_1) and the cubic negative damping term (ε) rise with the ascending reduced wind speed but the aerodynamic stiffness term (Y_2) declines while the negative damping term related to turbulence (ε_t) keeps stable. The rising turbulence intensity induces an uptrend for ε_t but a downtrend for ε and Y_1 , whereas Y_2 seems to be steady. Another point that can be found from these figures is

Fig. 14. Y_1 in the VIF model during 20:00 to 20:30.Fig. 15. Y_2 in the VIF model during 20:00 to 20:30.Fig. 16. ε in the VIF model during 20:00 to 20:30.

that the dispersion of ε and ε_t is more obvious than that of Y_1 and Y_2 . This is because the formers contribute less in VIV of the prototype bridge.

Based on the results discussed above, the relationships between the VIF parameters and wind characteristics (reduced wind speed (U_r) and turbulence intensity (I_w)) can be established. The wind attack angle and wind direction are not regarded as characteristics of the wind field because the wind field on site is transformed into the coordinate system of the deck already. In the establishment of the relationships, only the characteristics with the P values that are smaller than 0.05 are believed to relate strongly with the parameters. Based on this criterion, Y_1 and Y_2 are determined only by the reduced wind speed, ε_t is determined only by the turbulence intensity, and ε is determined by both reduced wind speed and turbulence intensity. The first order functions are adopted to establish the relationships and the least square method is employed to find their parameters. The established relationships can be expressed by Eq. (9) to Eq. (12).

$$Y_1 = 2.6087U_r - 17.852 \quad (9)$$

$$Y_2 = -1.1082U_r + 5.6261 \quad (10)$$

$$\varepsilon = 217.24U_r + 1190.8I_w - 1703.4 \quad (11)$$

$$\varepsilon_t = 1363.9I_w + 57.251 \quad (12)$$

5.3. Validation of the digital twin

By using the established relationships of VIF parameters in the last subsection, the numerical model has been converted to the digital twin which can be used to predict VIV of the prototype bridge under the action of turbulent winds. To validate its capability, the VIV-2 event, based on which the digital twin has been established, is predicted minute by minute by using the digital twin in this subsection. What shall be emphasized here is that the initial state of the digital twin in each prediction is set as the same as the bridge state measured on site. This is in line with the concept of digital twin, collaborating the physical and virtual entities, and this makes the digital twin can predict the time

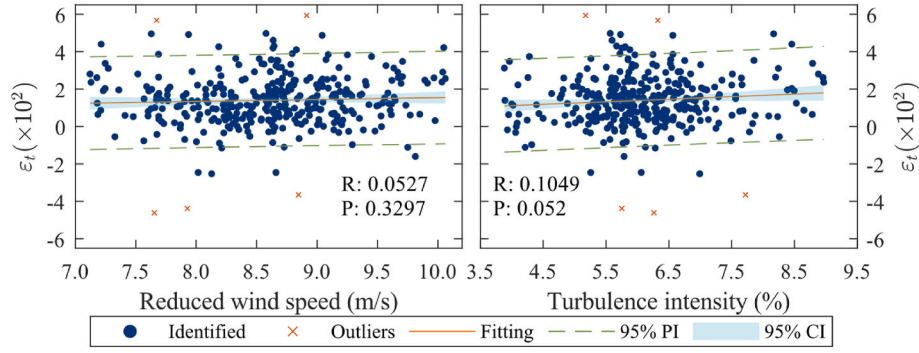


Fig. 17. ε_t in the VIF model during 20:00 to 20:30.

series of VIV rather than its amplitude only as the most traditional methods concern.

Fig. 18 shows the acceleration response time histories predicted by the digital twin and measured ones for the VIV-2 event, whereas Fig. 19 shows their Fourier amplitude spectra. Since the measured acceleration responses contain the buffeting-induced responses in higher frequency domain, a low-pass filter of 0.6 Hz is applied to the measured acceleration response time histories to obtain the filtered measured acceleration responses, which are also plotted in the figures. Fig. 20 to Fig. 22 contrast the acceleration responses and Fourier amplitude spectra from the measurement and the digital twin in the three selected minutes: 20:05, 20:15, and 20:25. From these figures, it can be concluded that the digital twin can predict VIV with a satisfying accuracy. The almost coincidental lines in the left figures and the sections ranging from 0 Hz to 0.5 Hz in the right figures prove that the digital twin succeeds in predicting VIV of the prototype bridge at the fundamental frequency. The differences between the predicted and measured acceleration responses appear at the frequencies higher than the fundamental frequency. The buffeting force-induced higher modal responses and the noises in the measured signals are attributed to such differences.

Considering that the vortex-induced peak acceleration response of the bridge is the key parameter for evaluating the comfort and functionality of the bridge and vehicles, the peak acceleration responses of the bridge deck are averaged within 1 min to obtain the 1-min average peak acceleration responses, to quantify the prediction accuracy. Table 5 lists the predicted, filtered, and measured 1-min average peak acceleration responses of the three time slots. The comparative results show

Table 4
Parameters in the fitting lines.

Term	For reduced wind speed		For turbulence intensity	
	Slope (k)	Intercept (b)	Slope (k)	Intercept (b)
Y_1	2.6078	-17.852	-39.926	7.0425
Y_2	-1.1082	5.6261	16.047	-4.9102
ε	206.69	-1540.3	-7178.8	671.64
ε_t	10.322	51.463	1363.9	57.251

that the predicted average peak acceleration responses are very close to the filtered measured average peak acceleration responses. The measured average peak acceleration responses are slightly larger than the predicted average peak acceleration responses.

6. Prediction of VIV by the digital twin

Providing better services for the reality are the target of a digital twin. In this section, the digital twin is employed to predict another VIV event to demonstrate the accuracy and capability of the digital twin for predicting the future VIV. The influence of the VIF model with the parameters obtained from wind tunnel tests on the prediction accuracy of the future VIV is also analyzed against the measurement and the prediction by the digital twin.

6.1. Prediction of VIV-1 event using the digital twin

The digital twin established in this study can be used to predict the acceleration response time histories of the prototype bridge for other VIV events. In this subsection, the VIV-1 event of the prototype bridge is predicted minute by minute. The VIV-1 event occurred at the natural

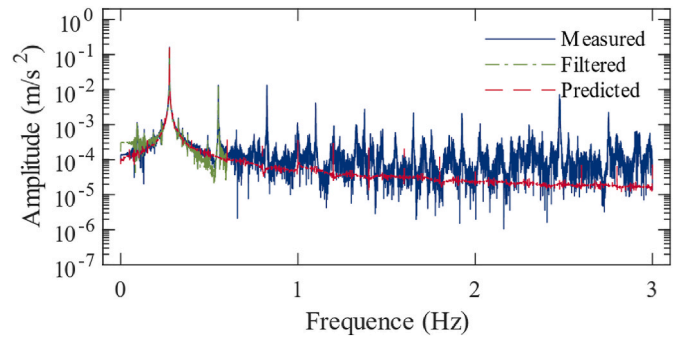


Fig. 19. Fourier amplitude spectra of acceleration responses predicted by the digital twin for VIV-2.

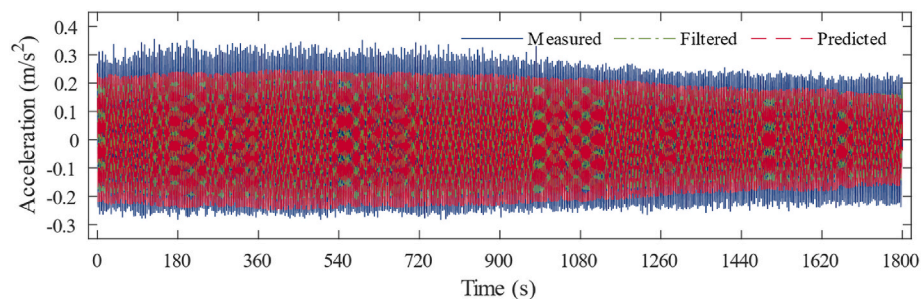


Fig. 18. Acceleration response time histories predicted by the digital twin for VIV-2.

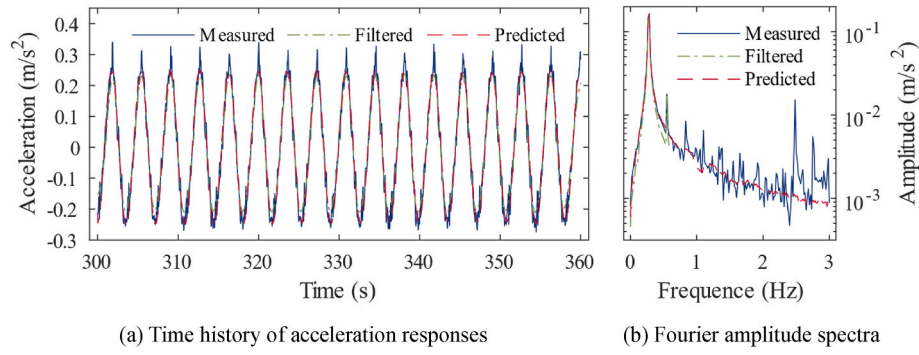


Fig. 20. Contrast the measured and predicted acceleration responses from 20:05 to 20:06.

frequency of 0.325 Hz, which is the sixth vertical mode of vibration, compared with the VIV-2 event, which occurred at the natural frequency of 0.274 Hz, which is the fifth vertical mode of vibration of the bridge. The mean wind speed and turbulence intensity during the VIV-1 event are also different from those during the VIV-2.

The initial state of the digital twin in the prediction is set as the same as the actual state of the prototype bridge measured on site and the wind field is transformed into the coordinate of the bridge deck. The contrasts between the measured and predicted acceleration responses as well as their Fourier amplitude spectra are shown in Fig. 23 to Fig. 27. These figures prove the accuracy of the digital twin in predicting another VIV event by the almost similar acceleration response time histories and Fourier amplitude spectra around the fundamental frequency at which VIV occurred.

As each VIV event occurred at only one vertical mode of vibration, the mode decomposition method is adopted and only the mode at which the VIV event occurred is considered in simulating the VIV event in this study. This means that only the vortex-induced acceleration response at one mode of vibration is simulated, and that the acceleration responses

Table 5

One-minute average peak acceleration responses during the 3 min in VIV-2.

	Measured (m/s ²)	Filtered (m/s ²)	Predicted (m/s ²)
20:05–20:06	0.3204	0.2313	0.2454
20:15–20:16	0.2839	0.2112	0.2103
20:25–20:26	0.2279	0.1694	0.1628

induced by the buffeting forces at all other modes of vibration, except the fundamental mode, are ignored although they do exist on the real bridge. This is the main reason why the predicted vortex-induced acceleration responses are lower than the measured ones, as shown in Fig. 23. This is also the reason why there are differences in the spectral amplitudes in high order vertical modes of vibration, as shown in Fig. 24.

Nevertheless, it is found from Fig. 25(a) that there is a small difference in the periods of the measured and predicted responses for Segment 1. Moreover, the level of agreement in Fig. 25 is lower than those in Fig. 26 and Fig. 27. The predicted 1-min average peak acceleration response of Segment 1 is also larger than the filtered measured one, as

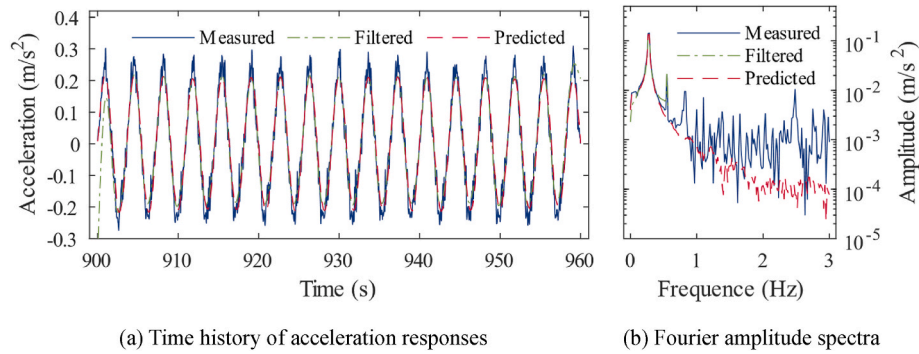


Fig. 21. Contrast the measured and predicted acceleration responses from 20:15 to 20:16.

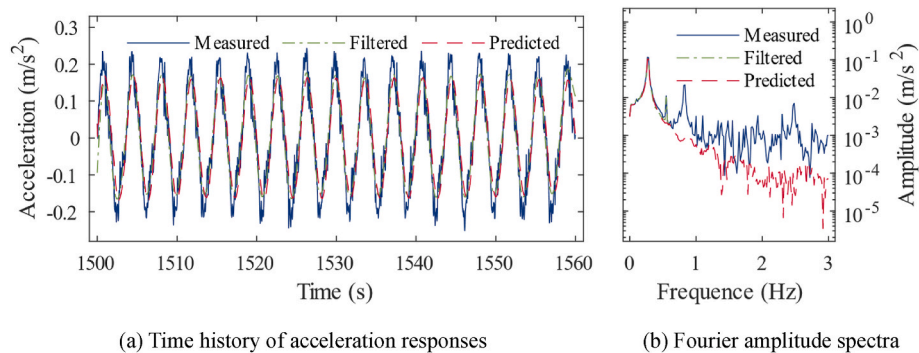


Fig. 22. Contrast the measured and predicted acceleration responses from 20:25 to 20:26.

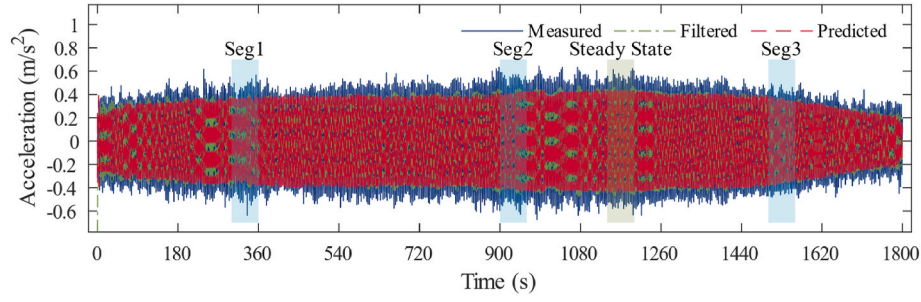


Fig. 23. Acceleration response time histories predicted by the digital twin for VIV-1.

shown in Table 6. One possible reason is that VIV occurrence during Segment 1 was not as steady as that during Segments 2 and 3 but the currently used VIF model is established mainly for the steady VIV. Therefore, developing a VIF model that can simulate the intact time histories of forces and responses of the deck throughout an entire VIV event is another important measure to be expected to improve the

accuracy of the digital twin.

6.2. Prediction of VIV-1 event using wind tunnel test results

Extensive wind tunnel tests of VIV of the deck section were conducted by Zhu et al. (2018), who identified the VIF parameters listed in Table 3. In order to compare the prediction accuracy of VIV-1 event between the digital twin and wind tunnel tests, all conditions were kept the same in subsequent analyses, except for the VIF parameters that were determined by the digital twin and wind tunnel tests, respectively. The initial state of the bridge was set to be the same in each prediction, matching the actual state measured on site, and the wind field was transformed into the coordinate system of the bridge deck.

The predicted acceleration responses time history and the Fourier amplitude spectrum for the VIV-1 event, based on the wind tunnel test results, are shown in Fig. 28 and Fig. 29, respectively, compared with the measured and filtered ones. It is seen from the two figures that the differences between the measurement and prediction are relatively large, particularly in the first half of the VIV-1 event where the predicted values are much larger than the measured values. This is because many uncertainties involved in the wind tunnel tests, compared with the real bridge and its wind environment, for determining the parameters in the

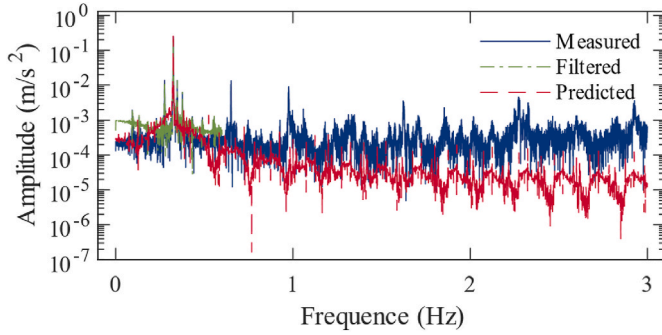


Fig. 24. Fourier amplitude spectra of acceleration responses predicted by the digital twin for VIV-1.

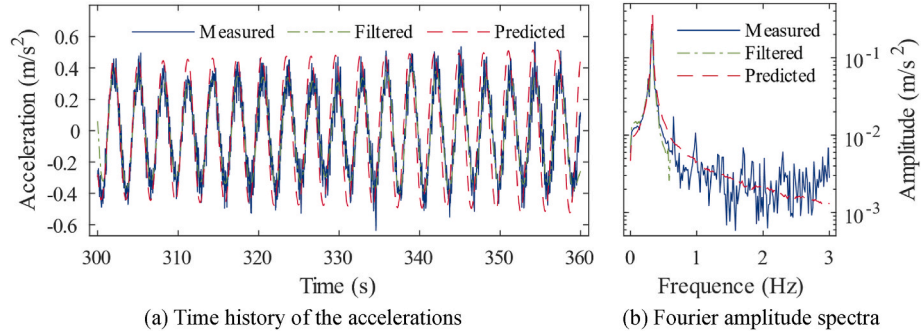


Fig. 25. Contrast the measured and digital twin predicted acceleration for Segment 1 at 19:05.

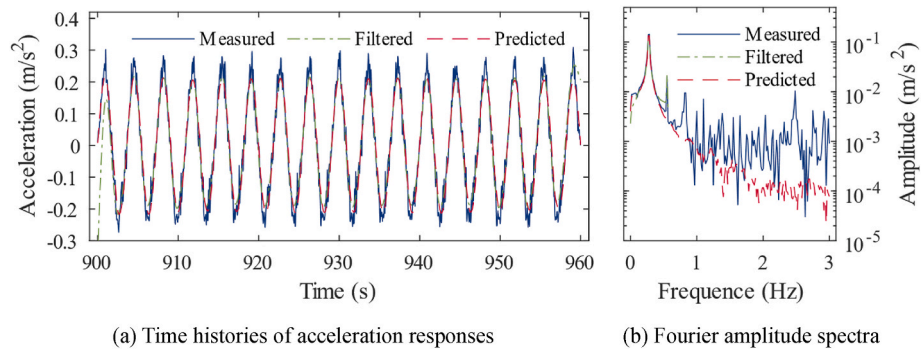


Fig. 26. Contrast the measured and digital twin predicted acceleration for Segment 2 at 19:15.

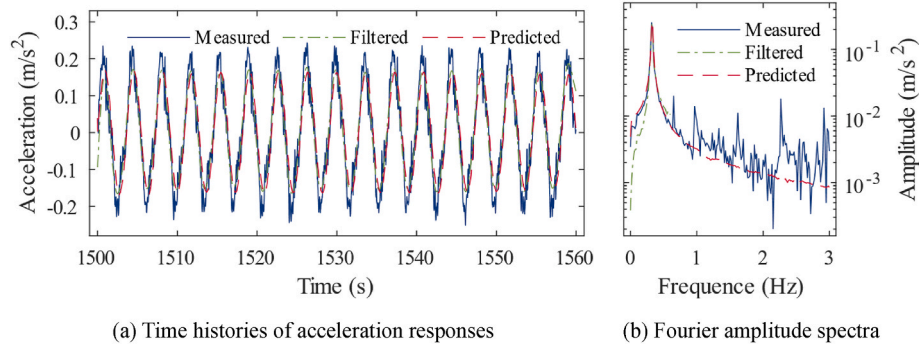


Fig. 27. Contrast the measured and digital twin predicted acceleration for Segment 3 at 19:25.

Table 6

One-minute average peak acceleration responses of the three segments in VIV-1.

	Measured (m/s^2)	Filtered (m/s^2)	Predicted (m/s^2)
Segment 1	0.4676	0.3473	0.4799
Segment 2	0.5400	0.4174	0.4562
Segment 3	0.4271	0.3455	0.3855

VIF model as explained below.

Fig. 30 shows the variations of the reduced wind speed and turbulence intensity with the time during the VIV-1 event. Fig. 31 displays the variations of the parameters in the VIF model, predicted by the digital twin (Eqs. (9)–(12)) and the wind tunnel test (linear interpolation), respectively, with the time. It is seen from Fig. 30 that the reduced wind speed and turbulence intensity vary with time. However, the turbulence intensity used in the wind tunnel tests for determining the parameters in the VIF model, as listed in Table 3, is constant. Establishing a digital twin to overcome the drawbacks of wind tunnel tests is thus necessary and essential for making VIV predictions.

Let us look at Fig. 31 closely. Before 600 s, the parameter Y_1 from the wind tunnel test is much larger than the value obtained by the digital twin. Since Y_1 represents the component of VIF in direct proportion to the bridge velocity (see Eq. (2)), its large value will cause large vibration. Therefore, the acceleration responses of the bridge in the first 600 s predicted by the wind tunnel test results are much larger than the measured ones and those predicted by the digital twin. Furthermore, different from the positive value of the parameter Y_2 obtained from the wind tunnel test, the digital twin gives the negative value of the parameter Y_2 , indicating that the displacement term in the VIF model given by the digital twin (see Eq. (2)) plays a restraining vibration role rather than an exciting vibration role. Although the correlation analyses show that the parameter ε is correlated to both reduced wind speed and turbulence intensity (see Fig. 16 and Eq. (11)), the almost overlap lines of the parameters ε shown in Fig. 31 indicate that the effect of turbulence intensity on this parameter is small. More importantly, while the parameter ε_t is constant in the wind tunnel tests (see Table 3), the results from the digital twin indicate that it positively correlates to the

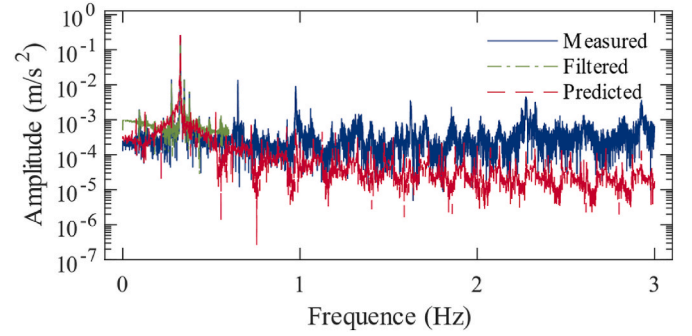


Fig. 29. Fourier amplitude spectra of acceleration responses predicted by the wind tunnel test for VIV-1.

turbulence intensity and can reach about two times over the wind tunnel test values. Based on the above discussions, the necessity of establishing a digital twin to overcome the drawbacks of wind tunnel tests for better simulating and predicting VIV events of the real bridge is further strengthened.

7. Concluding remarks

The framework and procedure for establishing a digital twin for simulating and predicting VIV of a long suspension bridge have been given in detail in this study. The accuracy of the digital twin for simulating and predicting VIV of a long suspension bridge has been assessed through a case study of the Xihoumen suspension bridge. The major conclusions and remarks from this study are summarized as follows:

- (1) The major components in the framework for establishing the digital twin include: (i) update the design document-based finite element (FE) model of the bridge against the measured dynamic characteristics; (ii) establish and update the numerical model for simulating VIV of the prototype bridge as a digital twin in terms of the VIF parameters and against the measured acceleration

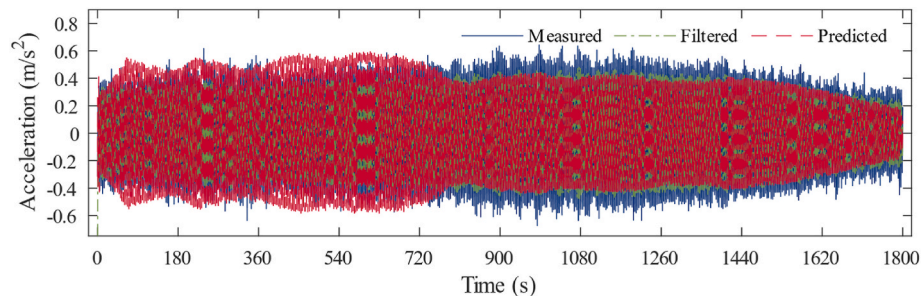


Fig. 28. Acceleration response time histories predicted by the wind tunnel test for VIV-1.

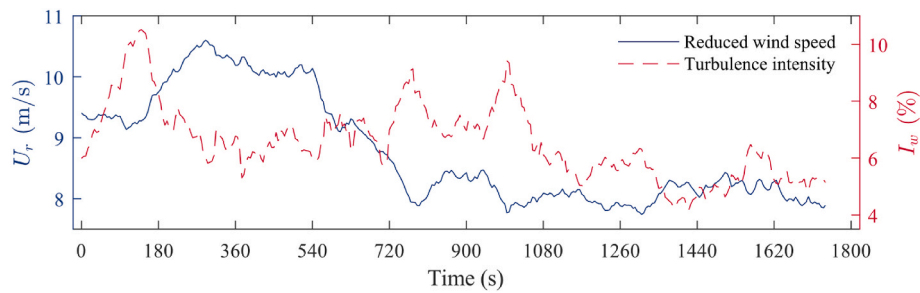


Fig. 30. Alteration of the reduced wind speed and turbulence intensity during VIV-1.

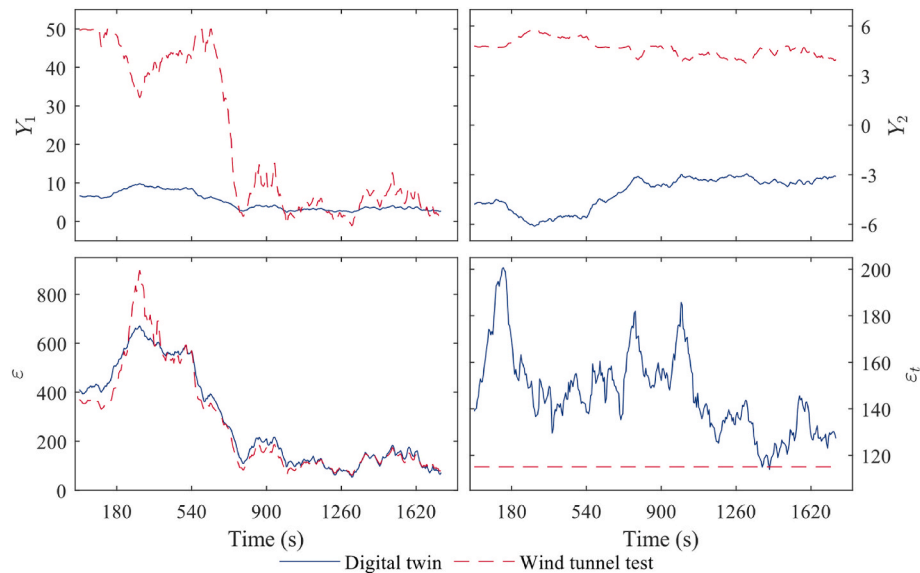


Fig. 31. Alteration of the VIF model parameters during VIV-1.

responses; and (iii) establish the relationships between the parameters in the VIF model and the wind characteristics to enable the digital twin to predict future VIV events.

- (2) The numerical model for simulating VIV of the prototype bridge is established by considering the updated FE model of the bridge and three types of wind forces: static wind forces, buffeting wind forces, and VIF.
- (3) The particle swarm optimization (PSO) method, used to identify (update) the VIF parameters in the numerical model against the measured acceleration responses of the bridge deck during the VIV events in this study, is a feasible method to find the best VIF parameters for an in-service bridge subjected to VIV.
- (4) For the in-service bridge used in the case study, the aerodynamic damping term (Y_1) and aerodynamic stiffness term (Y_2) in the VIF model are controlled mainly by the reduced wind speed while the negative damping term related to turbulence (ϵ_t) is dominated mainly by the turbulence intensity. The cubic negative damping term (ϵ) is, however, related to both reduced wind speed and turbulence intensity.
- (5) The digital twin established succeeds in simulating VIV of the prototype bridge at the fundamental dominant frequency. The differences between the simulated and measured acceleration responses appear at the frequencies higher than the fundamental frequency. The buffeting force-induced higher modal responses and the noises in the measured signals are attributed to such differences.
- (6) The digital twin established succeeds in predicting VIV of the prototype bridge at another dominant frequency and with different reduced wind speed and turbulence intensity. The

differences between the predicted and measured acceleration responses are also attributed to the buffeting force-induced higher modal responses and the noises existing in the measured signals.

- (7) The VIF parameters of the real bridge could be different from those obtained from the wind tunnel tests. Establishing a digital twin to overcome the drawbacks of wind tunnel tests is necessary and essential for better simulating and predicting VIV of the real bridge.

CRediT authorship contribution statement

Guo-Qing Zhang: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **You-Lin Xu:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing. **Dan-Hui Dan:** Data curation, Investigation. **Shang-Jun Jiang:** Investigation, Software. **Qing Zhu:** Data curation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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